



Suez University
Faculty of Petroleum and Mining Engineering
BSE225, Spring Term 16-17



Binary Arithmetic

Lecture 3 – Monday March 6, 2017

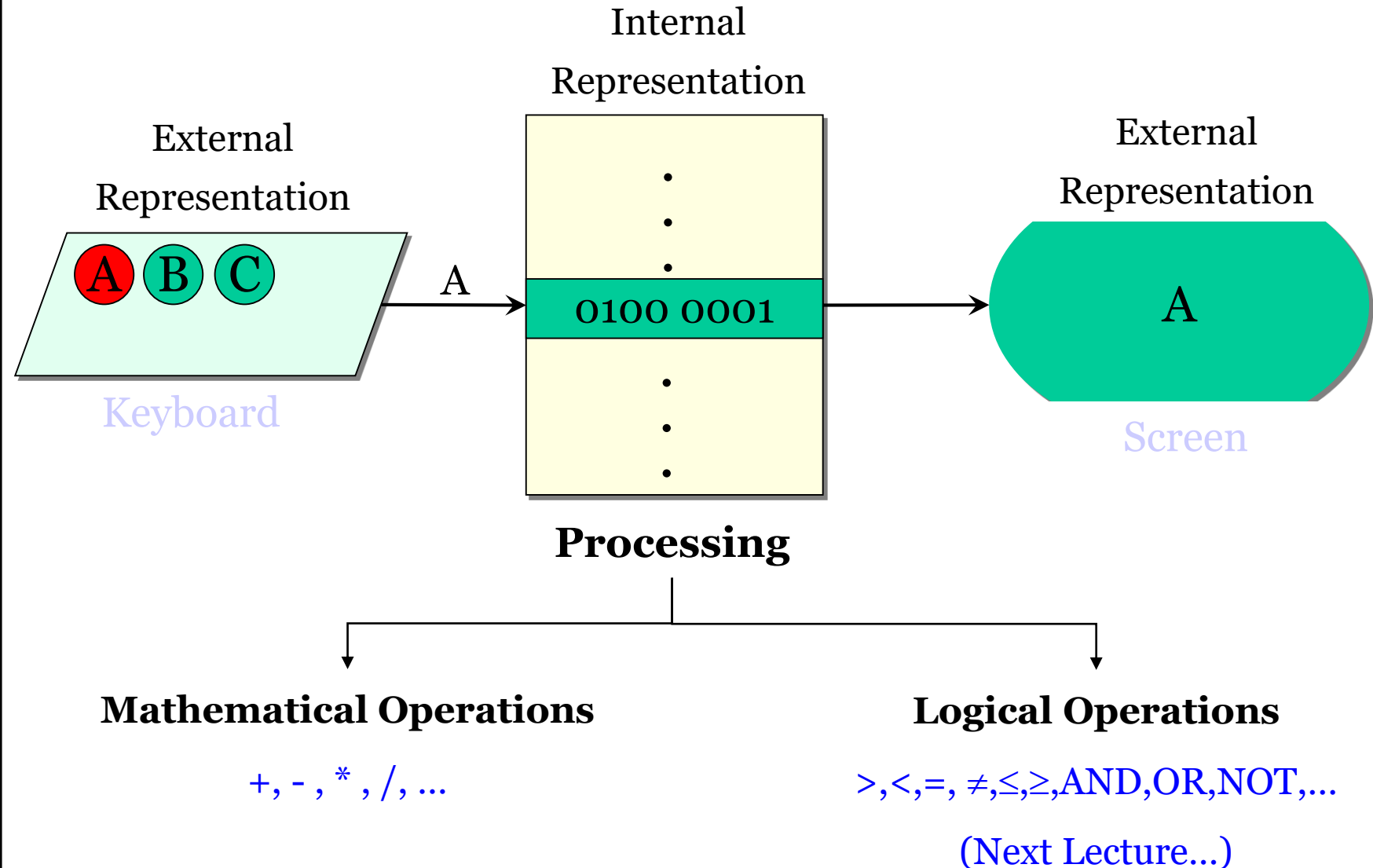
Outline

- Data Processing
- Unsigned Addition
- Representing Negatives
 - ◊ Signed-magnitude
 - ◊ 1's Complement
 - ◊ Two's Complement
- Binary Subtraction
- Binary Multiplication
- Binary Division
- Summary

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Data Processing



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Unsigned Addition

- The method of binary addition is the same as that for decimal addition.
- add the digits column-wise from the right to left.
- If the sum is greater than 1, a **carry** into the next column (to the next more significant bit) is generated.

Basic Binary Additions

				1
0	0	1	1	+1
+0	+1	+0	+1	+1
0	1	1	10	11

Unsigned Addition

- Decimal Addition and Binary Addition:

Decimal

1 Carries

0 6

+ 0 7

1 3

Unsigned Binary

1 1 Carries

0 1 1 0

+ 0 1 1 1

1 1 0 1

Check the result: $1101_2 = 13_{10}$



Unsigned Addition

Example-1: Perform the following addition in binary system:

$$00110_2 + 01111_2$$

Solution:

$$\begin{array}{r} 111 \text{Carries} \\ 00110 \\ + 01111 \\ \hline 10101 \end{array}$$

Check the result:

$$\begin{aligned} 00110_2 &= 6_{10} \\ 01111_2 &= 15_{10} \\ 10101_2 &= 21_{10} \end{aligned}$$

$$6_{10} + 15_{10} = 21_{10}$$


Unsigned Addition

Example-2: Perform the following addition in binary system:

$$0\ 1\ 1\ 011_2 + 0\ 1\ 1\ 010_2$$

Solution:

$$\begin{array}{rcccccc} & 1 & 1 & 0 & 1 & & \text{Carries} \\ & 0 & 1 & 1 & 0 & 1 & 1 \\ + & 0 & 1 & 1 & 0 & 1 & 0 \\ \hline 1 & 1 & 0 & 1 & 0 & 1 & \end{array}$$

Unsigned Addition

Example-3: Perform the following addition in binary system:

$$101.1001_2 + 0.1100_2$$

Solution:

$$\begin{array}{r}
 \text{Carries} \\
 0 \ 1 \ 1 \\
 1 \ 0 \ 1 \ . \ 1 \ 0 \ 0 \ 1 \\
 + 0 \ 0 \ 0 \ . \ 1 \ 1 \ 0 \ 0 \\
 \hline
 1 \ 1 \ 0 \ . \ 0 \ 1 \ 0 \ 1
 \end{array}$$

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Representing Negatives

- Singed-magnitude Representation.
- 1's Complement.
- 2's Complement.

Signed-magnitude

- In computers numbers are represented in fixed-size fields.
- Binary numbers can be used to represent positive and negative (signed) Integers.
- The **leftmost bit** is considered the sign bit.
- A zero in the sign bit indicates a positive number, and a one indicates a negative number.

Signed-magnitude Representation in an 8-bit Computer	
+12	-12
0000 1100	1000 1100

Signed-magnitude

- Signed-magnitude Representation:

Positive Numbers		Negative Numbers	
Decimal	Binary	Decimal	Binary
+0	0000 0000	-0	1000 0000
+1	0000 0001	-1	1000 0001
+2	0000 0010	-2	1000 0010
+3	0000 0011	-3	1000 0011
+4	0000 0100	-4	1000 0100
+5	0000 0101	-5	1000 0101
+6	0000 0110	-6	1000 0110
+7	0000 0111	-7	1000 0111
...	...	..	...



Problem:

The value 0 has two representations

1's Complement

- Reverse the bits to get the negative

Example: $0\ 1\ 0\ 1\ 0\ 1\ 1\ 1_2 = 87_{10}$

$1\ 0\ 1\ 0\ 1\ 0\ 0\ 0_2 = -87_{10}$

Positive Numbers		Negative Numbers	
Decimal	Binary	Decimal	Binary
+0	0000 0000	-0	1111 1111
+1	0000 0001	-1	1111 1110
+2	0000 0010	-2	1111 1101
+3	0000 0011	-3	1111 1100
+4	0000 0100	-4	1111 1011
...	...	..	...

The same problem:
 The value 0 has two representations

Two's Complement

- Invert each 1 to a 0 and each 0 to a 1 (1's complement)
- Add 1 to the LSB (rightmost bit) of the 1's complement.

Example-1: $01010111_2 = 87_{10}$

0 1 0 1 0 1 1 1

Original Number

1 0 1 0 1 0 0 0

Step-1: Invert

+

1

Step-2: Add 1

1 0 1 0 1 0 0 1

This is the two's complement, or -87_{10}

$$10101001_2 = -87_{10}$$

Two's Complement

Example-2: $11111011_2 = -5_{10}$

1 1 1 1 1 0 1 1

Original Number -5

0 0 0 0 0 1 0 0

Step-1: Invert

+

1

Step-2: Add 1

0 0 0 0 0 1 0 1

$00000101_2 = +5_{10}$

Two's Complement

Example-3: $00000000_2 = +0_{10}$ (use 8 bits for number representation).

0 0 0 0 0 0 0 0

Original Number +0

1 1 1 1 1 1 1 1

Step-1: Invert

+ 1

Step-2: Add 1

1 0 0 0 0 0 0 0

↑
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$$00000000_2 = +0_{10}$$

$$00000000_2 = -0_{10}$$

No Problem: The value 0 has the same representations

Two's Complement

Positive Numbers		Negative Numbers	
Decimal	Binary	Decimal	Binary
+0	0000 0000	-0	0000 0000
+1	0000 0001	-1	1111 1111
+2	0000 0010	-2	1111 1110
+3	0000 0011	-3	1111 1101
+4	0000 0100	-4	1111 1100
+5	0000 0101	-5	1111 1011
+6	0000 0110	-6	1111 1110
+7	0000 0111	-7	1111 1001
...	...	..	...

No Problem:

 The value 0 has the same representations

Note that computer will be able to distinguish between positive and negative numbers by looking at the leftmost bit of the number (if it's zero, the number is positive and if it's one, the number is negative)

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Binary Subtraction

- Convert the second number to two's complement and perform the binary addition:

$$A - B = A + (-B)$$

Use Signed Addition to do subtraction

Example-1: Perform the following subtraction in binary system: $010001_2 - 001010_2$
(use 6 bits for number representation).

Solution: $010001_2 - 001010_2 = 010001_2 + (-001010_2)$



Two's complement

Binary Subtraction

Example-1: Solution

$$010001_2 - 001010_2 = 010001_2 + (-001010_2)$$

Two's complement

- Complement: $110101 + 000001 = 110110$
- Addition: 1 Carries

$$\begin{array}{r}
 \text{Carries} \\
 1 \\
 0 \ 1 \ 0 \ 0 \ 0 \ 1 \\
 + 1 \ 1 \ 0 \ 1 \ 1 \ 0 \\
 \hline
 1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \\
 \uparrow \\
 \text{cut off this 1}
 \end{array}
 \quad
 010001_2 -$$

$$010001_2 - 001010_2 = 000111_2$$

Binary Subtraction

Example-1: Check the Result

$$010001_2 = 17_{10}$$

$$001010_2 = 10_{10}$$

$$000111_2 = 7_{10}$$

$$010001_2 - 001010_2 = 000111_2$$

$$17_{10} - 10_{10} = 7_{10} \quad \checkmark$$

Binary Subtraction

Example-2: Perform the following subtraction in binary system: $-6_{10} - 13_{10} = -19_{10}$
(use 6 bits for number representation).

Solution:

$$-6_{10} - 13_{10} = (-6_{10}) + (-13_{10})$$

$$6_{10} = 000110_2$$

$$-6_{10} = 111001 + 000001 = 111010_2$$

$$13_{10} = 001101_2$$

$$-13_{10} = 110010 + 000001 = 110011_2$$

Binary Subtraction

Example-2: Solution

$$-6_{10} - 13_{10} = (-6_{10}) + (-13_{10})$$

$$-6_{10} = 111010_2$$

$$-13_{10} = 110011_2$$

1 0 0 1

Carries

1 1 1 0 1 0

+ 1 1 0 0 1 1

1 1 0 1 1 0 1



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$$-6_{10} - 13_{10} = 101101_2$$

Binary Subtraction

Example-2: Check the Result

$$-6_{10} - 13_{10} = -19_{10}$$

$$19_{10} = 010011_2$$

$$-19_{10} = 101100 + 000001 = 101101_2$$

Result of binary
subtraction is

101101_2



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Binary Multiplication

- Decimal Multiplication

239_{10} Multiplicand

$\times 123_{10}$ Multiplier

717_{10} 1st partial product

478_{10} 2nd partial product

$+ 239_{10}$ 3rd partial product

$29,397$ Final product

Binary Multiplication

- Binary Multiplication using partial products

1010011 ₂	Multiplicand	1110010001 ₂	Sum
× 0111011 ₂	Multiplier	+1010011 ₂	5 th product
1010011 ₂	1 st partial product	100011000001 ₂	Sum
+1010011 ₂	2 nd partial product	+1010011 ₂	6 th product
11111001 ₂	Sum of 1 st and 2 nd	1001100100001 ₂	Sum
+0000000 ₂	3 rd partial product	+0000000 ₂	7 th product
011111001 ₂	Sum	1001100100001 ₂	Final
+1010011 ₂	4 th partial product		
1110010001 ₂	Sum		

Binary Multiplication

- Binary Multiplication using partial products

$$\begin{array}{r} 1\ 0\ 1\ 0\ 0\ 1\ 1_2 \text{ Multiplicand} \\ \times 0\ 1\ 1\ 1\ 0\ 1\ 1_2 \text{ Multiplier} \\ \hline 1\ 0\ 1\ 0\ 0\ 1\ 1_2 \text{ 1st partial product} \end{array}$$

Since the sign of the final product is a 1, determined in step 1, take the 2'd complement of the product

$$1\ 0\ 0\ 1\ 1\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 1_2 \quad \Rightarrow \quad 0\ 1\ 1\ 0\ 0\ 1\ 1\ 0\ 1\ 1\ 1\ 1\ 1_2$$

Attach the sign bit

$$1\ 0\ 1\ 1\ 0\ 0\ 1\ 1\ 0\ 1\ 1\ 1\ 1\ 1_2$$

Exercise: verify the multiplication is correct by converting to decimal numbers and performing the multiplication.

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Binary Division

- Decimal division

$$\frac{\text{dividend}}{\text{divisor}} = \text{quotient}$$

21_{10}

dividend

$- 7_{10}$

1st subtraction of divisor

14_{10}

1st partial remainder

$- 7_{10}$

2nd subtraction of divisor

7_{10}

2nd partial remainder

$- 7_{10}$

3rd subtraction of divisor

0_{10}

Zero reminder

Divisor was
subtracted from the
dividend three times

Quotient=3

Binary Division

- Binary division

$$\begin{array}{r} 01100100 \\ \hline 00011001 \end{array} = ?$$

Step 1: The signs of both numbers are positive, so the quotient will be positive. The quotient is initially zero: 00000000

Step 2: Subtract the divisor from the dividend using 2' complement addition.

01100100	dividend
+11100111	2's complement of divisor
01001011	Positive 1 st partial remainder

Add 1 to quotient: 00000000 + 00000001 = 00000001

Binary Division

- Binary division

Step 3: Subtract the divisor from the 1st partial remainder using 2's complement addition:

01001011	1 st partial remainder
+11100111	2's complement of divisor
00110010	Positive 2 nd partial remainder

Add 1 to quotient: 00000001 + 00000001 = 00000010

Binary Division

- Binary division

Step 4: Subtract the divisor from the 2nd partial remainder using 2's complement addition:

00110010	2 nd partial remainder
+11100111	2's complement of divisor
00011001	Positive 3 rd partial remainder

Add 1 to quotient: $00000010 + 00000001 = 00000011$

Binary Division

- Binary division

Step 5: Subtract the divisor from the 3rd partial remainder using 2's complement addition:

00011001	3 rd partial remainder
+11100111	2's complement of divisor
00000000	Positive 3 rd partial remainder

Add 1 to quotient: $00000011 + 00000001 = 00000100$ (**final quotient**)

Exercise: verify the division is correct by converting to decimal numbers and performing the division.

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Summary

- Arithmetic is at the heart of the digital computer, and the majority of arithmetic performed by computers is binary arithmetic.
- The method of unsigned addition in binary system is the same as that for decimal addition.
- Adding signed numbers is not significantly different from adding unsigned numbers. Only you should convert the signed number to two's complement and perform the binary addition.
- Computers operate on numbers of fixed size.